Handwritten Digit Recognition

Comparison of MLP CNN and LSTM Architectures

Interim report on

Deep Learning Project [ICT-4442]

Submitted By

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Individual Project



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PART B INTERIM REPORT

Team No. and Names: Individual project (no team) | YASH SHUKLA | 210911380

Title: Handwritten Digit Recognition - Comparison of MLP, CNN and LSTM Architectures

GitHub Repository Link: https://github.com/yashshukla27/Handwritten-Digit-Recognition/upload

1. Literature Review

Ten papers are compared below. Published results use their own datasets and protocols; they cannot be directly compared with the small 8 x 8 dataset used here.

Paper (Author, Year)	Method	Dataset	Key Result	Relevance to Project
[1] LeCun et al. (1989)	Task-specific backpropagation network	USPS ZIP-code digits	1% error with about 9% rejection; not ordinary full-coverage accuracy.	Motivates spatial structure; our CNN has no rejection option.
[2] Ciresan et al. (2010)	Deep MLP and deformed training images	MNIST	0.35% error reported with large networks and GPU training.	Supports an MLP baseline; our small model uses no augmentation.
[3] Ciresan et al. (2012)	Multi-column deep CNN ensemble	MNIST; traffic signs and other benchmarks	Averaging separately trained columns improved benchmark recognition.	Historical high-performing approach; larger training budget than our single CNN.
[4] Hochreiter and Schmidhuber (1997)	LSTM memory cells and gates	Synthetic long-lag sequence tasks	Learned dependencies exceeding 1,000 time steps.	Foundation for proposed row-wise LSTM; not a digit-accuracy benchmark.
[5] Le, Jaitly and Hinton (2015)	Identity-initialized ReLU RNN	Sequential MNIST; adding task; language; TIMIT	3% sequential-MNIST error vs 34% for their lightly tuned LSTM.	Shows image-to-sequence evaluation is valid; our 8 rows are much shorter than 784 pixels.
[6] Kingma and Ba (2014/2015)	Adam adaptive optimizer	MNIST; CIFAR-10; text classification	Efficient first-order optimization using moving gradient moments.	Same Adam rule used for both models; does not guarantee best accuracy.
[7] Srivastava et al. (2014)	Dropout regularization	Vision, speech, document and biological benchmarks	Reduced overfitting by randomly disabling training units.	Candidate final-phase ablation; not enabled in interim models.
[8] Ioffe and Szegedy (2015)	Batch normalization	ImageNet classification	Matched baseline accuracy with 14 times fewer training steps in reported experiment.	Potential optimization extension; not implemented in this small-data baseline.
[9] He et al. (2015/2016)	Residual CNNs	ImageNet; CIFAR-10; COCO	152-layer models; ensemble ImageNet test top-5 error 3.57%.	Explains deeper image architectures; excessive scope for this interim.

[10] Hinton, Vinyals and Dean (2015)	Knowledge distillation	MNIST; speech recognition	Smaller MNIST network: 146 errors directly trained, 74 with distillation.	Possible MLP-from-CNN extension; our MLP currently learns from labels only.

Synthesis: MLPs provide a dense baseline [2], CNNs encode local spatial structure [1,3], and LSTMs model ordered information [4,5]. Adam supports optimization [6]. Dropout, normalization, residual learning and distillation [7]-[10] suggest future extensions. This is an educational comparison, not a new architecture or a claim of current state-of-the-art performance.

2. Dataset Acquisition & Preprocessing

The objective is to compare three architecture families for ten-class recognition of isolated handwritten digits. The scikit-learn load_digits dataset contains 1,797 grayscale images of size 8 x 8, labels 0-9, and intensities 0-16. It is not MNIST. It is distributed with scikit-learn, so no separate image download is needed.

Pipeline stage	Implementation
Cleaning and checks	Validate shape, finite values, ten labels and non-overlapping splits. No examples removed; no missing values found.
Normalization and encoding	Convert pixels to float64 and divide by the known maximum 16. Labels remain integer class IDs 0-9.
Augmentation	None at interim. No resizing or external images.
Train / validation / test	Stratified split with seed 42: 1,257 / 270 / 270 samples. Both models reuse saved indices.
Model inputs	MLP: 64-value vector. CNN: 8 x 8 grid. Proposed LSTM: eight rows, each with eight features.

3. Models Implemented So Far

Model	Owner (Member)	Status	Preliminary Metric	Notes
MLP	YASH SHUKLA	Implemented; assisted run verified	Test accuracy 96.67%	64 > 64 ReLU > 32 ReLU > 10; 6,570 parameters.
CNN	YASH SHUKLA	Implemented; assisted run verified	Test accuracy 97.78%	3 x 3 valid convolution, 8 filters > ReLU > flatten 288 > Dense 32 ReLU > 10; 9,658 parameters.
LSTM	YASH SHUKLA	Planned for final	Pending	8 rows > LSTM hidden size 32 > Dense 10.

Common setup: cross-entropy loss; Adam learning rate 0.001, beta1 0.9, beta2 0.999, epsilon 1e-8; batch size 64; 40 epochs; seed 42. The first checkpoint with the highest validation accuracy is selected, then evaluated on the test set. No test-based tuning. Python 3.12.14, NumPy 2.3.5, scikit-learn 1.8.0, CPU.

Model	Best epoch	Val acc %	Test acc %	Macro precision	Macro recall	Macro F1
MLP	28	95.93	96.67	0.9682	0.9664	0.9667
CNN	33	97.04	97.78	0.9786	0.9773	0.9776

MLP correctly classified 261/270 test images; CNN classified 264/270. The difference is three images, or 1.11 percentage points. CNN filters may help capture strokes, but parameter counts differ and only one seed was run. Macro metrics average

the ten class scores equally. Saved predictions and weights reproduce the metrics; sampled gradient checks passed with errors below 5e-11.

Error analysis: the six CNN mistakes were 2 to 1; 8 to 1 twice; 1 to 6; 6 to 8; and 8 to 5. Small or unusual digit shapes may be ambiguous. The dataset is small, the split is not writer-independent, and performance on phone photographs is untested. Evidence is included in results/ and the runnable notebook.

4. Individual Contribution Log (to date)

Sole student and model owner: YASH SHUKLA, registration number 210911380. Ownership identifies responsibility for the project; the completed work below explicitly distinguishes AI-assisted output from student verification.

Member Name	Reg. No.	Task(s) Completed	Signature
YASH SHUKLA	210911380	Provided individual-project details, registration number, GitHub account and signature. AI-assisted package includes literature summaries, preprocessing, MLP/CNN code, verified experiments and report. Personal code review, rerun and Git commits remain to be recorded.	Signature reproduced below.

Signature of YASH SHUKLA



Signature image supplied by the student. Git commit history: pending; no past commits or independent student implementation are claimed.

5. Risk / Plan for Remaining Work

Target date 2026	Remaining work and evidence
3-4 October	Review and rerun the notebook; record actual individual work; confirm the revised interim deadline; create the repository under yashshukla27 and push genuine commits.
5-10 October	Implement row-wise LSTM; verify input shapes; train and evaluate with the shared split and common metrics.
11-17 October	Repeat all models with three seeds; tune using validation only; report mean and standard deviation.
18-25 October	Expand to 12+ references, analyze errors, prepare Part C and practise the 10-12 minute presentation.
26-29 October; 31 October	Final presentation during 26-29 October. Submit final report and repository link by 31 October, 11:59 PM.

Risks and mitigation: small-data overfitting - keep compact models and repeat seeds; LSTM training difficulty - validate shapes and tune on validation data; inconsistent comparisons - reuse splits; installation problems - test the LSTM framework early; limited time as a solo student - prioritize the required third architecture and a reproducible comparison.

AI-use disclosure: ChatGPT (OpenAI), accessed 3 October 2026 India time, assisted with topic selection, literature summaries, code, gradient checks, experiment execution, charts, notebook and report drafting. Results were generated on 2 October UTC / 3 October India time in the

assistant environment. They do not show student work before 25 September. Personal review, edits and reruns should be recorded with their actual dates.

Submission checks pending: real repository URL and commit history; confirmation of the actual interim deadline; institutional Turnitin similarity and AI-content reports (guideline limits: 15% plagiarism and 20% AI content). No detector score is certified.

References

[1] LeCun, Y., Boser, B. E., Denker, J. S., Henderson, D., Howard, R. E., Hubbard, W. E., and Jackel, L. D. Handwritten Digit Recognition with a Back-Propagation Network. NIPS 2, 1989. https://papers.neurips.cc/paper_files/paper/1989/hash/53c3bce66e43be4f209556518c2fcb54-Abstract.html

[2] Ciresan, D. C., Meier, U., Gambardella, L. M., and Schmidhuber, J. Deep Big Simple Neural Nets Excel on Handwritten Digit Recognition. Neural Computation 22, 2010. https://arxiv.org/abs/1003.0358

[3] Ciresan, D., Meier, U., and Schmidhuber, J. Multi-column Deep Neural Networks for Image Classification. CVPR, 2012, pp. 36423649. https://arxiv.org/abs/1202.2745

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[5] Le, Q. V., Jaitly, N., and Hinton, G. E. A Simple Way to Initialize Recurrent Networks of Rectified Linear Units. arXiv, 2015. https://arxiv.org/abs/1504.00941

[6] Kingma, D. P., and Ba, J. Adam: A Method for Stochastic Optimization. arXiv 2014; ICLR 2015. https://arxiv.org/abs/1412.6980

[7] Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., and Salakhutdinov, R. Dropout: A Simple Way to Prevent Neural Networks from Overfitting. JMLR 15, 2014, pp. 1929-1958. https://jmlr.org/papers/v15/srivastava14a.html

[8] Ioffe, S., and Szegedy, C. Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift. ICML, 2015. https://arxiv.org/abs/1502.03167

[9] He, K., Zhang, X., Ren, S., and Sun, J. Deep Residual Learning for Image Recognition. arXiv 2015; CVPR 2016. https://arxiv.org/abs/1512.03385

[10] Hinton, G., Vinyals, O., and Dean, J. Distilling the Knowledge in a Neural Network. NIPS Deep Learning and Representation Learning Workshop, 2015. https://arxiv.org/abs/1503.02531

Dataset: Scikit-learn, load_digits documentation. https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_digits.html. Accessed 3 October 2026.

Course documents: ICT 4442 Mini Project Guidelines and ICT 4442 Mini Project Report Template, supplied by the student. This report follows Part B with the provided institutional cover and a single-student contribution row.

Reproducibility files: train.py; check_gradients.py; Run_Project.ipynb; results/metrics.json; results/split_indices.npz; saved model weights, predictions, confusion matrices and run logs.